

# Dave's Daily

AI • Bayesian optimization • US equities

Altman and Amodei told the UN Security Council to coordinate on AI risk a day after Trump dismissed “globalist” controls; Anthropic’s Claude mined a CRISPR-like enzyme system; and hot PMI costs plus Barr’s hawkishness re-anchored October hike odds. Fresh arXiv BO work tracks moving optima with active inference and folds LLMs into auxiliary-informed search.

## Markets

The mechanism today was a *policy-path* shock, not a single print. [CNBC](#) reports S&P Global’s flash PMI showed manufacturing and services at multi-year highs while input-cost inflation jumped to its steepest pace since late 2022—fuel, transport, and wages feeding through. Hours later, Fed Governor Michael Barr said that in his base case “further policy adjustments are likely to be needed” after last week’s first hike in three years. CME FedWatch odds for an October move jumped into the low seventies; the front end of the Treasury curve led the reprice. That is the substance for equity risk: a higher path for short rates raises discount rates on long-duration cash flows—the same AI-capex stories that had been carrying semis and platforms—while oil’s rebound removes the soft-landing cushion traders had been leaning on. [Kiplinger](#) and the [AP](#) both frame the selloff as the bond market asserting itself. Relative to yesterday’s Warsh “feel the economy” read, Barr’s remarks pull the committee back toward a data-driven tightening bias—two voices, one book of futures.

## AI

At an emergency UN Security Council session on Wednesday, [OpenAI’s Sam Altman](#) and [Anthropic’s Dario Amodei](#) urged international coordination on frontier risk. Altman: “Beating companies in a competitive race is not a reason to make rash decisions”; Amodei pledged to “slow down as much as necessary” for safe successive releases and argued no nation can manage the technology alone. The politics collide with [Tuesday’s UNGA speech](#), where Trump rejected what he called a “globalist scheme” to control AI and vowed not to rein in growth. Society’s stake is institutional capacity: labs asking for shared brakes while the host government treats restraint as a competitive own-goal—and while both firms still face IPO-timing questions about what “slow” means commercially.

On the research side, Anthropic launched a life-sciences lab and reported that Claude agents—with high-level human direction—discovered [array-associated reverse transcriptases \(ART\)](#): a phage RT plus a CRISPR-like DNA-repeat array and an accessory protein, spotted after roughly 950 agents and 210 million tokens of genome mining. Feng Zhang called it a genuine AI-assisted biological find. Separately, Google DeepMind detailed [Private AI Compute](#) with encrypted server-side memory: keys stay on-device, envelopes open only in hardware enclaves—an attempt to give assistants cross-device continuity without surrendering on-device privacy norms.

## Bayesian optimization

An arXiv title search this evening turns up several BO submissions dated 22–16 September. Three that speak to scale, sample efficiency, and AI coupling: [BOBA](#) (Kelly, Patel, Thomas, and colleagues; 22 Sep) designs an acquisition for *dynamic* BO inspired by active-inference free energy. Standard UCB-style DBO under-allocates probes when  $f(x, t)$  drifts; BOBA adds forward-looking uncertainty about how the function will change, improving regret when query budgets are tight. That is scaling BO into non-stationary, real-time loops.

[Bayesian Optimization with Rich Auxiliary Information via LLMs](#) (Gupta, Karagözlü, Sonker, Póczos, and Schnieder; 16 Sep) treats training curves, notes, and images as first-class signals. Three LLM-mediated ways to inject that context beat plain BO and prior LLM optimizers on HPO and a nuclear-fusion control task—a direct weld of language models to the BO loop.

[Nearly-Instant Latent Space BO](#) (Fan, Doumont, Kalisz, and colleagues; 16 Sep) exploits linear surrogates on spherical latents so acquisition and fitting are nearly closed form— $\geq 100\times$  faster than strong baselines while matching molecular and image generative-design screens. When virtual evaluations are cheap, wall-clock—not sample count—was the bottleneck; this removes it.

Updated 23 September 2026 evening. Market narrative emphasizes policy path and duration risk, not index prints. Created with [malvin](#).